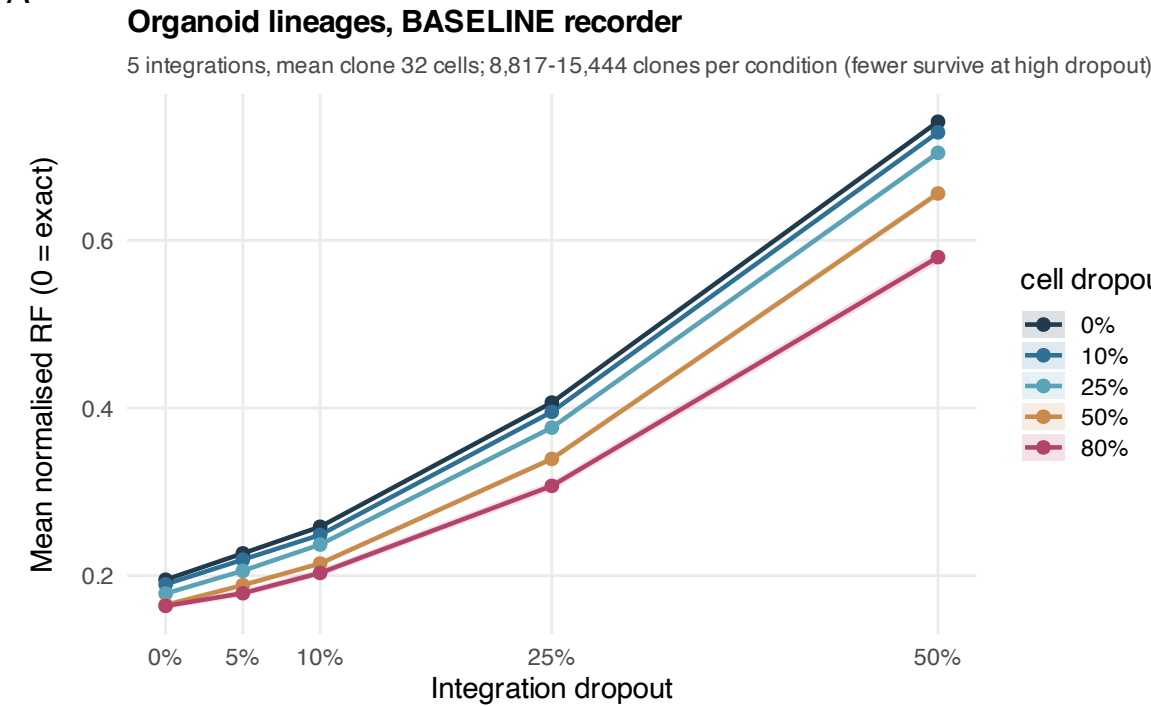


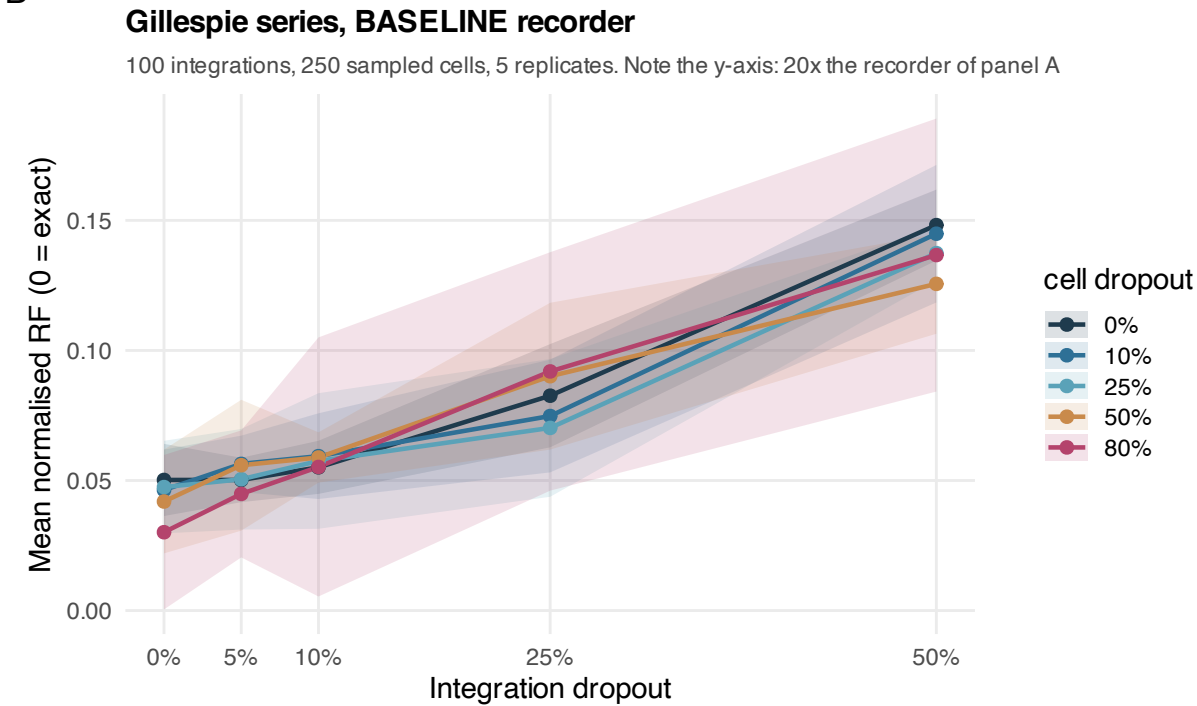
Reconstruction accuracy under integration and cell dropout

Integration dropout makes characters missing in cells that remain; cell dropout removes tips and the truth tree is pruned to match.

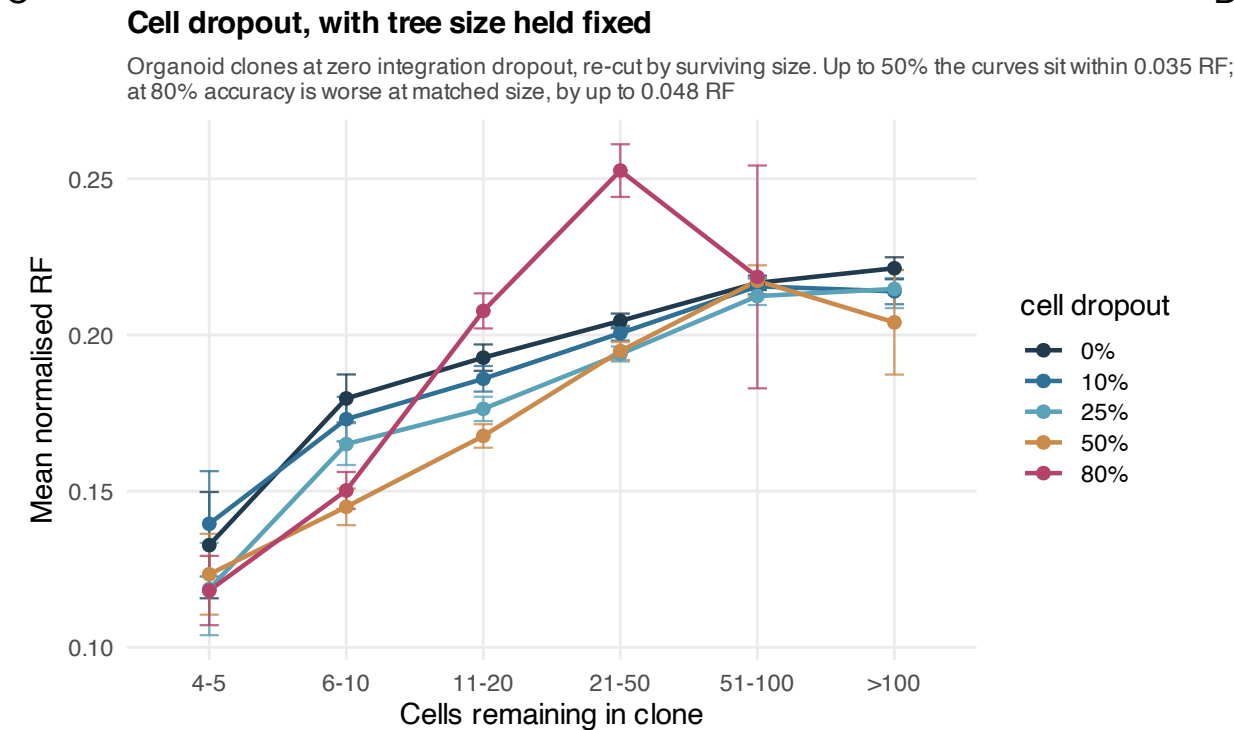
A



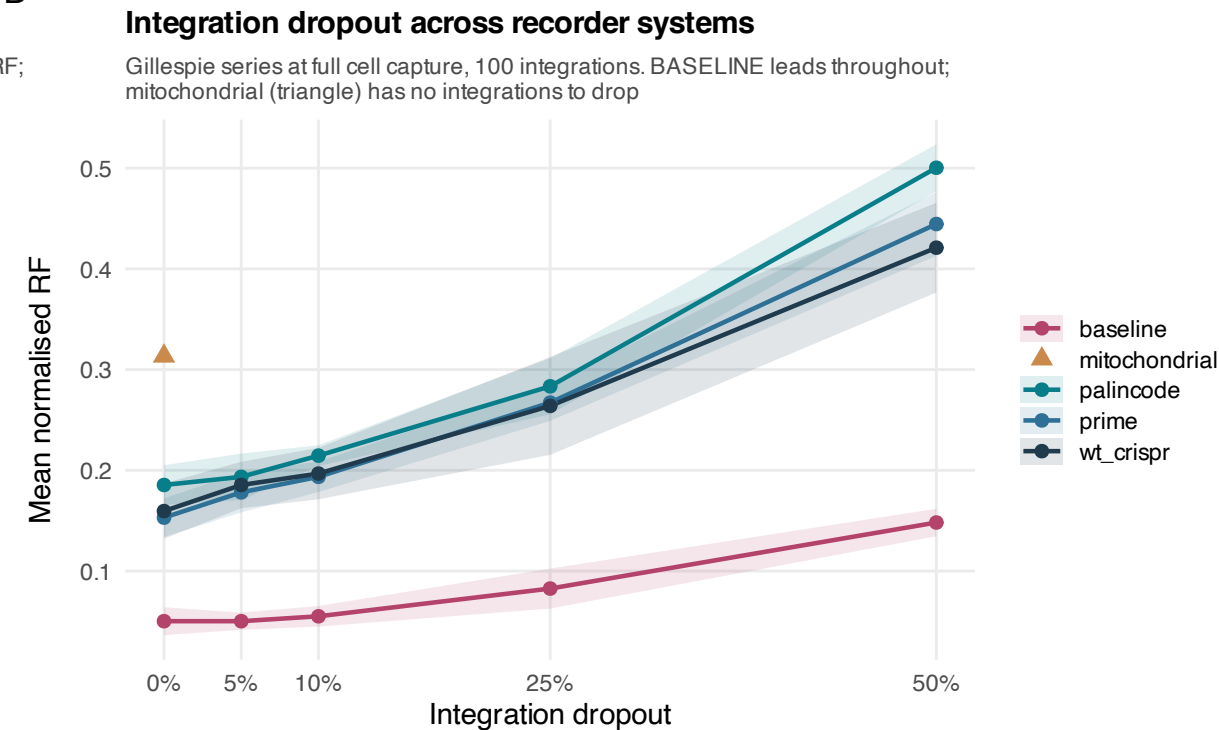
B



C



D



Falling RF along the cell-dropout axis in A and B is mostly a size effect, not a recovery effect: normalised RF is a fraction of the internal splits of the surviving tree, and dropping cells leaves fewer splits to get wrong -- mean organoid clone size falls from 32.4 cells to 10.0 across that axis. Panel C removes it by re-cutting on surviving size, and the improvement largely disappears; at 80% dropout accuracy is worse at matched size. Note that matched surviving size is not a matched clone: a 21-50 cell tree at 80% dropout came from a clone of roughly 105-250 cells, so those bins are drawn from larger, older clones. Integration dropout, which removes characters without removing tips, is the axis that costs accuracy unambiguously.